

Semi-Automatic Tracking Pipeline for Small Aerial Objects

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Output Video Explanation

Overview:

The output video demonstrates a semi-automatic tracking pipeline applied to a small aerial object (skydiver). A bounding box is used to track the object across frames. The system allows manual initialization and re-initialization to ensure accurate tracking.

Pipeline Logic:

- The object is manually selected in the first frame using ROI selection.
- A CSRT tracker is initialized using this bounding box.
- The tracker follows the object across subsequent frames.
- If tracking drifts or fails, the user can reinitialize the tracker.

CSRT Tracker:

CSRT (Discriminative Correlation Filter with Channel and Spatial Reliability) is a visual tracking algorithm that models the appearance of the object and updates it over time. It is robust to moderate scale changes and partial occlusions.

Tracking Performance:

The system achieved a **tracking rate of 98.62%**. However, this metric alone is not fully reliable, as the tracker may continue tracking an incorrect region without signaling failure. Detecting an object in a highly visual noise environment like this video is proven to be difficult with the data provided. Therefore, manual reinitialization was used to maintain accuracy.

Human-in-the-Loop Correction:

The tracker was manually reinitialized **4 times** during the video to correct drift and ensure accurate tracking.

Approach & Iterative Development

Initial Approach (Detection-Based):

We began with a detection-based approach using a YOLOv11s model. Due to the extremely small size of the object and low visual contrast, the model achieved a **tracking rate of approximately 4%**.

Motion-Based Approach (Optical Flow):

We then explored optical flow methods available in OpenCV to track motion patterns. While this improved performance, background motion and noise significantly affected accuracy, resulting in a tracking rate of approximately **20%**.

Final Approach (Semi-Automatic Tracking):

The final solution uses a **manual ROI + CSRT tracking pipeline**. This approach prioritizes robustness over full automation:

- Manual initialization ensures correct object identification.
- CSRT provides stable frame-to-frame tracking. This helped us to get seamless tracking for hundreds of frames without any human input.
- Manual reinitialization corrects drift when necessary.

Automation Attempts:

We attempted to automate object detection using classical computer vision techniques. However, these methods struggled due to the low signal-to-noise ratio and small object size.

Limitations & Future Improvements

Challenges Identified:

- The object occupies a very small portion of the frame.
- Low contrast between the object and background.
- Non-uniform camera motion.
- Scene transitions and abrupt visual changes.

Limitations of RGB Data:

The current approach relies solely on RGB visual data, which is insufficient for this scenario. The object's appearance blends with the background, making reliable detection difficult.

Potential Improvements:

- **Multi-modal data:** Incorporating thermal or infrared data to improve object visibility.
- **Improved detection models:** Training deep learning models specifically for small object detection.
- **Hybrid pipelines:** Combining detection and tracking methods for better robustness.
- **Camera setup:** Using stabilized or zoomed-in video to improve signal quality.

Summary:

This work highlights the importance of balancing automation with robustness when dealing with challenging real-world scenarios involving small object tracking. This pipeline is directly extensible to drone and UAV detection, the same challenges of small object size, camera motion, and low contrast apply to counter-UAS systems.